Smaran Rangarajan Bharadwaj

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EDUCATION

- **RV University, Bengaluru** · B.Tech (Hons.), Computer Science and Engineering — *Graduating: Aug 2026 · CGPA: 9.80*
- **Sri Kumaran Children's Home, Bengaluru** · Class 12 (CBSE) - PCMC — *2022 · Score: 95.2%*

TECHNICAL SKILLS

Languages: Python3, Go, C, C++, Java, JavaScript, YAML
Developer Tools: Docker, Kubernetes, Git, AWS, GCP, Azure, Jira, Confluence, HuggingFace
Databases: MySQL, PostgreSQL, SQLite, MongoDB, ClickHouse
Frameworks: TensorFlow, PyTorch, Node.js, Flask, Streamlit, gRPC, RESTful APIs, Ollama, Kafka, Flink, Unsloth, LangChain
Relevant Coursework: Google Cloud Computing Foundations, AWS Networking Masterclass, Big Data Computing

EXPERIENCE

Software Development Intern — *Jan 2026 - Jul 2026*
Aviatrix Systems Pvt. Ltd. — *Bengaluru*

- Designed and shipped the scalable, cloud-native Data-Bridge backend in Go using gRPC and ClickHouse materialized views, enabling sub-second queries across 10M+ records.
- Architected and shipped a distributed multi-cloud (AWS/GCP) risk assessment ETL pipeline leveraging Flink and ClickHouse analyzing 22 resource types in under 100 seconds.
- Automated CI/CD workflows for the Data-Bridge service with GitHub Actions, reducing deployment time by 68%.

Project Intern — *Aug 2025 - Jan 2026*
Aeronautical Development Establishment (DRDO) — *Bengaluru*

- Architected a multi-stage mission debriefing pipeline processing 10,000+ flight parameters across 6 missions in under 15 minutes with parallelized processing.
- Engineered an event detection engine using Allen's Interval Algebra and expert systems to classify overlapping anomalies.

ML Engineer Intern — *Jun 2025 - Aug 2025*
GovernAIce — *Berkeley (Remote)*

- Designed an end-to-end multi-lingual RAG pipeline reducing compliance report generation time by 73% over manual processes by leveraging DeepSeek-R1:70B and MongoDB Atlas Vector Search.
- Developed a policy ingestion pipeline supporting 6 languages and 12,000+ legal documents utilizing IBM Docling and XLM-RoBERTa embeddings.

LEADERSHIP AND AWARDS

- Founded and established RV University's IEEE Student Branch, growing it from 5 founding ExeCom members to 60+ members across 3 society chapters and 20+ events.
- Awarded merit scholarships for 3 consecutive years, ranking 2nd among the university's top academic performers.
- Won 2nd place among 117 teams at the HPCC Safe Havens Hackathon for developing a distributed computing solution for refugee support systems.
- Co-designed and published a patent for Gelos, a fitness-focused interactive game, alongside Dr. Phani Kumar Pullela.

PROJECTS

Braille GPT — *Multi-agent System, LoRA LLM Fine Tuning*

- Developed a multi-agent system built on LoRA-fine-tuned LLaMA 3.1 8B with a Streamlit-based front end, converting text and image inputs into bilingual Braille with a 0.76 average BLEU score across translation agents.

Obstacle Avoidance for Visually Impaired — *Computer Vision, YOLO, Transfer Learning*

- Led a four-member team to develop a YOLOv8n-based obstacle detection system for the visually impaired, achieving 82.4% mAP on a self collected dataset and 132ms average inference latency.

MITM Attack Simulator — *Network Security, Flask, PostgreSQL*

- Engineered a full-stack network security toolkit processing 15,000+ packets per second, using Scapy and Flask to simulate MITM attacks across a multi-VM bridged network.
- Secured API endpoints and session audit logging across 3 user roles, by architecting a PostgreSQL database with JWT authentication.